

Abdul Rehman ☺

Multivariable Calculus

BS-CS-Sem-II

Partial Derivative

Ex 16.3, Swokowski, page 732:-

Question 1-18, Find first partial derivative

Question no 1:-

$$f(x, y) = 2x^4y^3 - xy^2 + 3y + 1$$

$$f_x(x, y) = 2y^3(4)x^3 - y^2$$

$$f_x(x, y) = 8x^3y^3 - y^2$$

$$f_y(x, y) = 2(x^4)(y^2)(3) - 2xy + 3$$

$$f_y(x, y) = 6x^4y^2 - 2xy + 3$$

$$f(x, y) = (x^3 - y^2)^2$$

$$f_x(x, y) = 2(x^3 - y^2)(3x^2)$$

$$f_x(x, y) = 6x^2(x^3 - y^2)$$

$$f_y(x, y) = 2(x^3 - y^2)(-2y)$$

$$f_y(x, y) = -4y(x^3 - y^2)$$

Question no 3:-

$$f(x, s) = \sqrt{x^2 + s^2}$$

$$f_x(x, s) = \frac{1(2x)}{2\sqrt{x^2 + s^2}} = \frac{x}{\sqrt{x^2 + s^2}}$$

$$f_s(x, s) = \frac{2s}{2\sqrt{x^2 + s^2}} = \frac{s}{\sqrt{x^2 + s^2}}$$

Question no 4:-

$$f(s, t) = t/s - s/t$$

$$\begin{aligned} f_s(s, t) &= \frac{t(-1)}{s^2} - \frac{1}{t} \\ &= -\frac{t}{s^2} - \frac{1}{t} \end{aligned}$$

$$\begin{aligned} f_t(s, t) &= \frac{1}{s} - \frac{s(-1)}{t^2} \\ &= \frac{1}{s} + \frac{s}{t^2} \end{aligned}$$

Question no 5:-

$$f(x, y) = xe^y + y \sin x$$

$$f_x(x, y) = e^y + y \cos x$$

$$f_y(x, y) = xe^y + \sin x$$

Question no 6:-

$$f(x, y) = e^x \ln xy$$

$$f_x(x, y) = \frac{e^x y}{xy} + e^x \ln xy = e^x \left(\frac{1}{x} + \ln xy \right)$$

$$f_y(x, y) = \frac{e^x y x}{xy} = \frac{e^x}{y}$$

Question no 7:-

$$f(t, v) = \ln \sqrt{(t+v)/(t-v)}$$

$$f_t(t, v) = \frac{\sqrt{(t-v)/(t+v)}}{2\sqrt{(t+v)/(t-v)}} \left[\frac{(t-v) - (t+v)}{(t-v)^2} \right]$$

$$f_t(t, v) = \frac{-v}{t^2 - v^2}$$

$$f_v(t, v) = \frac{\sqrt{(t-v)/(t+v)}}{2\sqrt{(t+v)/(t-v)}} \frac{(t-v) + (t+v)}{(t-v)^2}$$

$$f_v(t, v) = \frac{t}{t^2 - v^2}$$

Question no 8:-

$$f(u, w) = \tan^{-1}(u/w)$$

$$\begin{aligned} f_u(u, w) &= \frac{1}{1 + \frac{u^2}{w^2}} \left(\frac{1}{w} \right) \\ &= \frac{w^2}{w^2 + u^2} \left(\frac{1}{w} \right) \\ &= \frac{w}{w^2 + u^2} \end{aligned}$$

$$f_w(u, w) = \frac{1}{1 + \frac{u^2}{w^2}} \cdot \left(\frac{-u}{w^2} \right)$$

$$f_w(u, w) = \frac{-u}{w^2 + u^2}$$

Question no 9:-

$$f(x, y) = x \cos(x/y)$$

$$f_y(x, y) = -x \sin\left(\frac{x}{y}\right) \left(\frac{-1}{y^2} \right) (x)$$

$$f_y(x, y) = \frac{x^2}{y^2} \sin\left(\frac{x}{y}\right)$$

$$f_x(x, y) = -x \sin\left(\frac{x}{y}\right) \left(\frac{1}{y} \right) + \cos\left(\frac{x}{y}\right)$$

$$f_x(x, y) = -\frac{x}{y} \sin\left(\frac{x}{y}\right) + \cos\left(\frac{x}{y}\right)$$

Question no 10:-

$$f(x, y) = \sqrt{4x^2 - y^2} \sec x$$

$$f_x(x, y) = \sqrt{4x^2 - y^2} \sec x \tan x + \frac{\sec x (8x)}{2\sqrt{4x^2 - y^2}}$$

$$f_y(x, y) = \sec x \left(\frac{1(-2y)}{2\sqrt{4x^2 - y^2}} \right)$$

$$f_y(x, y) = \frac{-y \sec x}{\sqrt{4x^2 - y^2}}$$

Question no 11:-

$$f(x, s, t) = x^2 e^{2s} \cos t$$

$$f_x(x, s, t) = 2 \cos t (x) (e^{2s})$$

$$f_x(x, s, t) = 2x e^{2s} \cos t$$

$$f_s(x, s, t) = x^2 \cos t e^{2s} (2)$$

$$f_s(x, s, t) = 2x^2 e^{2s} \cos t$$

$$f_t(x, s, t) = -x^2 e^{2s} \sin t$$

Question no 12:-

$$f(x, y, t) = \frac{(x^2 - t^2)}{(1 + \sin 3y)}$$

$$f_x(x, y, t) = \frac{(2x)}{(1 + \sin 3y)}$$

$$f_y(x, y, t) = \frac{+3(x^2 - t^2) \cos 3y}{(1 + \sin 3y)^2}$$

$$f_{yy}(x, y, t) = \frac{-3 \cos 3y (x^2 - t^2)}{(1 + \sin 3y)^3}$$

$$f_t(x, y, t) = \frac{-2t}{1 + \sin 3y}$$

Question no 13:-

$$F(x, y, z) = (y^2 + z^2)^n$$

$$F_x(x, y, z) = \frac{\partial}{\partial x} (y^2 + z^2)^n$$

$$\frac{\partial}{\partial x} \ln F(x, y, z) = n \ln(y^2 + z^2)$$

$$F_x(x, y, z) = \ln(y^2 + z^2) (y^2 + z^2)^n$$

$$F_y(x, y, z) = n (y^2 + z^2)^{n-1} (2y)$$

$$F_y(x, y, z) = 2ny (y^2 + z^2)^{n-1}$$

$$F_z(x, y, z) = 2nz (y^2 + z^2)^{n-1}$$

Question no 14:-

$$F(x, s, v) = (2x + 3s)^{\cos v}$$

$$F_x(x, s, v) = \cos v (2x + 3s)^{\cos v - 1} (2)$$

$$F_x(x, s, v) = 2 \cos v (2x + 3s)^{\cos v - 1}$$

$$F_s(x, s, v) = 3 \cos v (2x + 3s)^{\cos v - 1}$$

$$F_v(x, s, v) = (2x + 3s)^{\cos v} (\cos v) (-\sin v) (2x + 3s)$$

$$F_v(x, s, v) = -\sin v \ln (2x + 3s) (2x + 3s)^{\cos v}$$

Question no 15:-

$$f(x, y, z) = xe^z - ye^x + ze^{-y}$$

$$f_x(x, y, z) = e^z - ye^x$$

$$f_y(x, y, z) = -e^x + ze^{-y}(-1)$$

$$f_y(x, y, z) = -e^x - ze^{-y}$$

$$f_z(x, y, z) = xe^z + e^{-y}$$

Question no 16:-

$$f(x, s, v, p) = x^3 \tan s + \sqrt{s} e^{x^2} - v \cos 2p$$

$$f_x(x, s, v, p) = 3x^2 \tan s$$

$$f_s(x, s, v, p) = x^3 + \frac{e^{x^2}}{2\sqrt{s}} + x^3 \sec^2(s)$$

$$f_p(x, s, v, p) = +v \sin 2p(2)$$

$$f_p(x, s, v, p) = 2v \sin 2p$$

$$f_v(x, s, v, p) = (x^3 \tan s + \sqrt{s} e^{x^2} - v \cos 2p) \left(\frac{1}{v} + 2v - 1 \right)$$

$$f_v(x, s, v, p) = (x^3 \tan s + \sqrt{s} e^{x^2} - v \cos 2p) (2v - 1)$$

$$f_v(x, s, v, p) = 2v \sqrt{s} e^{x^2} - \cos 2p$$

Question no 17:-

$$f(q, v, w) = \sin^{-1} \sqrt{qv} + \sin vw$$

$$f_v(q, v, w) = \frac{1}{2\sqrt{1-qv}(qv)} + \cos vw(w)(1)$$

$$f_q(q, v, w) = \frac{v}{2\sqrt{(1+qv)(qv)}}$$

$$f_v(q, v, w) = \frac{q}{2\sqrt{(1+q,v)(q,v)}} + w \cos vw$$

$$f_w(q, v, w) = v \cos vw$$

Question no 18:-

$$f(x, y, z) = xyz e^{xyz}$$

$$f_x(x, y, z) = yz(x e^{xyz}(yz) + e^{xyz})$$

$$f_x(x, y, z) = yz(xyz e^{xyz} + e^{xyz})$$

$$f_y(x, y, z) = xz(y e^{xyz}(xz) + e^{xyz})$$

$$f_z(x, y, z) = xy(z e^{xyz}(xy) + e^{xyz})$$

Question no 19:-

$$W = xy^4 - 2x^2y^3 + 4x^2 - 3y$$

$$W_x = y^4 - 4xy^3 + 8x$$

$$W_{xy} = +4y^3 + 8(0) + (-4x(3)(y^2)) = 4y^3 - 12xy^2$$

$$W_y = 4xy^3 - 6x^2y^2 - 3$$

$$W_{yx} = 4y^3 - 12xy^2$$

Hence proved,

$$W_{xy} = W_{yx}$$

Question no 20:-

$$W = x^2 / (x+y)$$

$$W_x = \frac{(x+y)(2x) + (-x^2)}{(x+y)^2}$$

$$W_x = \frac{3x^2 + 2xy - x^2}{(x+y)^2}$$

$$W_x = \frac{2x^2 - x^2 + 2xy}{(x+y)^2}$$

$$W_x = \frac{x^2 + 2xy}{(x+y)^2}$$

$$W_{xy} = \frac{(x+y)^2(2x) - (x^2 + 2xy)(2)(x+y)}{(x+y)^2(x+y)^2}$$

$$W_{xy} = \frac{2(x+y)(x+y)(x) - x^2 - 2xy}{(x+y)^4}$$

$$W_{xy} = \frac{2(x+y)(-xy)}{(x+y)^4}$$

$$W_y = \frac{(x+y)(0) - (x^2)(1)}{(x+y)^2}$$

$$W_y = \frac{-x^2}{(x+y)^2}$$

$$W_{yx} = \frac{(x+y)^2(-2x) - (-x^2)(2)(x+y)}{(x+y)^4}$$

$$W_{yx} = \frac{(x+y)(2)(-x^2 - xy + x^2)}{(x+y)^4}$$

$$W_{yx} = \frac{-2xy(x+y)}{(x+y)^4}$$

Proved, $W_{xy} = W_{yx}$

Question no 21:-

$$W = x^3 e^{-2y} + y^{-2} \cos x$$

$$W_x = 3x^2 e^{-2y} - y^{-2} \sin x$$

$$W_{xy} = -6x^2 e^{-2y} + 2y^{-3} \sin x$$

$$W_y = -2x^3 e^{-2y} - 2y^{-3} \cos x$$

$$W_{yx} = -6x^2 e^{-2y} + 2y^{-3} \sin x$$

Proved,

$$\underline{W_{xy} = W_{yx}}$$

Question no 22:-

$$W = y^2 e^{x^2} + (1/x^2 y^3)$$

$$W_x = (y^2 e^{x^2} + (1/x^2 y^3)) (2x - \frac{2y^3}{x^3})$$

$$W_x = (y^2 e^{x^2} + \frac{1}{x^2 y^3}) (\frac{2x^2 - 2}{x})$$

$$W_{xy} = (\frac{2ye^{x^2}}{x^2 y^4} - \frac{3}{x^2 y^4}) (\frac{2x^2 - 2}{x})$$

$$W_y = \frac{2ye^{x^2}}{x^2 y^4} - \frac{3}{x^2 y^4}$$

$$W_{yx} = (\frac{2ye^{x^2}}{y^4 x^2} - \frac{3}{y^4 x^2}) (\frac{2x - \frac{2xy^4}{x^2}}{x})$$

$$W_{yx} = (\frac{2ye^{x^2}}{x^2 y^4} - \frac{3}{x^2 y^4}) (\frac{2x^2 - 2}{x})$$

Proved,

$$\underline{W_{xy} = W_{yx}}$$

Question no 23:-

$$W = x^2 \cosh(z/y)$$

$$W_x = 2x \cosh\left(\frac{z}{y}\right)$$

$$W_{xy} = 2xz \sinh\left(\frac{z}{y}\right)$$

$$W_y = x^2 z \sinh\left(\frac{z}{y}\right)$$

$$W_{yx} = 2xz \sinh\left(\frac{z}{y}\right)$$

Proved,

$$\underline{W_{yx} = W_{xy}}$$

Question no 24:-

$$W = \sqrt{x^2 + y^2 + z^2}$$

$$W_x = \frac{2x}{2\sqrt{x^2 + y^2 + z^2}}$$

$$W_{xy} = \frac{-x(2y)}{2(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

$$W_{yx} = \frac{-xy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

$$W_y = \frac{2y}{2\sqrt{x^2 + y^2 + z^2}}$$

$$W_{yz} = \frac{-2xy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

$$W_{yx} = \frac{-xy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

Question no 25:-

If $W = 3x^2y^3z + 2xy^4z^2 - yz$, find W_{xyz}

$$W = 3x^2y^3z + 2xy^4z^2 - yz$$

$$W_x = 6xy^3z + 2y^4z^2 - y$$

$$W_{xy} = 18xy^2z + 8y^3z^2$$

$$W_{xyz} = 18xy^2 + 16y^3z$$

Question no 26:-

$W = u^4vt^2 - 3uv^2t^3$, find W_{tut}

$$W = u^4vt^2 - 3uv^2t^3$$

$$W_t = 2u^4vt - 9uv^2t^2$$

$$W_{tu} = 8u^3vt - 9v^2t^2$$

$$W_{tut} = 8u^3v - 18v^2t$$

Question no 27:-

$u = v \sec rt$, find u_{rrv}

$$u_r = vt \sec rt \tan rt$$

$$u_{rv} = t \sec rt \cdot \tan rt$$

$$u_{rrv} = t(\sec rt \cdot \sec^2 rt + \sec rt \tan^2 rt)$$

$$u_{rr} = \underline{\underline{t \sec r (\sec^2 r + \tan^2 r)}}$$

Question no 28:-

If $V = y \ln(x^2 + z^4)$, find V_{zz}

$$V = y \ln(x^2 + z^4)$$

$$V_z = \frac{4yz^3}{x^2 + z^4}$$

$$V_{zz} = \frac{4y((x^2 + z^4)(3z^2) - z^3(4z^3))}{(x^2 + z^4)^2}$$

$$V_{zz} = \frac{4yz^2\{(3)(x^2 + z^4) - (z^4)(4)\}}{(x^2 + z^4)^2}$$

$$V_{zz} = \frac{12x^2yz^2 - 4yz^6}{(x^2 + z^4)^2}$$

$$V_{zz} = \frac{12x^2yz^2 - 4yz^6}{(x^2 + z^4)^2} = \frac{4z^2(3x^2 - z^4)}{(x^2 + z^4)^2}$$

Question no 29:- $\frac{\partial W}{\partial x}$

$W = \sin xyz$, find $\frac{\partial W}{\partial z} \frac{\partial W}{\partial y} \frac{\partial W}{\partial x}$

$$\frac{\partial W}{\partial z} = xy \cos xyz$$

$$\frac{\partial^2 W}{\partial z^2}$$

$$\frac{\partial^2 W}{\partial z^2} = x(-y x z \sin xyz + \cos xyz)$$

$$\frac{\partial^2 W}{\partial z \partial y}$$

$$\frac{\partial^3 W}{\partial z \partial y \partial x} = (x \sin xyz (yz) + \cos xyz) - (yz)(yz x^2 \cos xyz + 2x \sin xyz)$$

$$\begin{aligned}
 &= -xyz \sin(xyz) + \cos(xyz) - x^2 y^2 z^2 \cos xyz \\
 &\quad - 2xyz \sin xyz \\
 &= \underline{\underline{(1 - x^2 y^2 z^2) \cos xyz - 3xyz \sin xyz}}
 \end{aligned}$$

Question no 30:-

$W = x^2/(y^2 + z^2)$, find $\partial^3 W / \partial z \partial y^2$

$$\frac{\partial W}{\partial z} = \frac{-x^2(2z)}{(y^2 + z^2)^2}$$

$$\frac{\partial W}{\partial z \partial y} = \frac{+4x^2 y z (z)}{(y^2 + z^2)^3}$$

$$\frac{\partial W}{\partial z \partial y^2} = \frac{8x^2 z \left(\frac{y}{3} \right) (y^2 + z^2)^3 - (y/3) (y^2 + z^2)^2 (2y)}{(y^2 + z^2)^6}$$

$$= \frac{8x^2 z (y^2 + z^2)^2 (y^2 + z^2 - 6y^2)}{(y^2 + z^2)^6}$$

$$= \frac{8x^2 z (z^2 - 5y^2)}{(y^2 + z^2)^4}$$

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Question no 31:-

$$W = x^4 s^3 t - 3s^2 e^{xt},$$

Verify that: $W_{xrs} = W_{sxs} = W_{sxs}$

$$W = x^4 s^3 t - 3s^2 e^{xt}$$

$$W_x = 4x^3 s^3 t - 3ts^2 e^{xt}$$

$$W_{xx} = 12x^2 s^3 t - 3t^2 s^2 e^{xt}$$

$$W_{xrs} = 36x^2 s^2 t - 6t^2 s e^{xt}$$

$$W_{rs} = 12x^3 s^2 t - 6t s e^{xt}$$

$$W_{xsr} = 36x^2 s^2 t - 6t^2 s e^{xt}$$

$$W_s = 3x^4 s^2 t - 6s e^{xt}$$

$$W_{sr} = 12x^3 s^2 t - 6t s e^{xt}$$

$$W_{rsr} = 36x^2 s^2 t - 6t^2 s e^{xt}$$

Proved,

$$\underline{W_{xrs} = W_{ysr} = W_{srr}}$$

Question no 32:-

$$W = \tan uv + 2 \ln(u+v)$$

Verify, $W_{uvv} = W_{vuv} = W_{vvu}$

$$W = \tan uv + 2 \ln(u+v)$$

$$W_u = \sec^2(vu) v + \frac{2}{u+v}$$

$$W_{uv} = \sec^2(vu) + 2v \sec^2(vu) \tan(vu) - 2(u+v)^{-2}$$

$$W_{uv} = 2uv \sec^2(uv) \tan(uv) + \sec^2(uv) - 2(u+v)^{-2}$$

$$W_{uvv} = 4u \sec^2(uv) \tan(uv) + 4u^2 v \sec^2(uv) \tan^2(uv) + 2u^2 v \sec^4(uv) + 4(u+v)^{-3}$$

$$W_v = u \sec^2(uv) + (u+v)^{-1} (v)$$

$$W_{vu} = 2uv \sec^2(uv) \tan(uv) - 2(u+v)^{-2} + \sec^2(uv)$$

$$W_{uvv} = 4u \sec^2(uv) \tan(uv) + 4u^2 v \sec^2(uv) \tan^2(uv) + 2u^2 v \sec^4(uv) + 4(u+v)^{-3}$$

$$W_{vv} = 2u^2 \sec^2(uv) \tan(uv) - 2(u+v)^{-2}$$

$$W_{vvu} = 4u \sec^2(uv) \tan(uv) + 4u^2 v \sec^2(uv) \tan^2(uv) + 2u^2 v \sec^4(uv) + 4(u+v)^{-3}$$

Proved,

$$W_{uvv} = W_{vvu} = W_{uvu}$$

Question 33 to 36, let $W = f(x, y)$,
prove that $W_{xx} + W_{yy} = 0$

Question no 33:-

$$f(x, y) = \ln \sqrt{x^2 + y^2}$$

$$f_x(x, y) = \frac{2x}{2(x^2 + y^2)}$$

$$f_{xx}(x, y) = \frac{2(-1)(x^2 + y^2)^{-2}(2x) + 2(x^2 + y^2)^{-1}}{(x^2 + y^2)^2}$$

$$f_{xx}(x, y) = (x^2 + y^2)^{-1} - 2x^2(x^2 + y^2)^{-2}$$

$$f_y(x, y) = \frac{y}{(x^2 + y^2)}$$

$$f_{yy}(x, y) = 2y^2(-1)(x^2 + y^2)^{-2} + (x^2 + y^2)^{-1}$$

$$f_{yy}(x, y) + f_{xx}(x, y) = \frac{1}{(x^2 + y^2)} - \frac{2x^2}{(x^2 + y^2)^2} + \frac{1}{(x^2 + y^2)} - \frac{2y^2}{(x^2 + y^2)^2}$$

$$= \frac{2y^2}{(x^2 + y^2)^2} + \frac{2x^2}{(x^2 + y^2)^2} = \frac{2(x^2 + y^2)}{(x^2 + y^2)^2} = \frac{2}{x^2 + y^2}$$

$$= \frac{2}{x^2+y^2} - \frac{2x^2 - 2y^2}{(x^2+y^2)^2}$$

$$= \frac{2}{x^2+y^2} \left(\frac{(x^2+y^2) - (x^2+y^2)}{(x^2+y^2)} \right)$$

$$= 0$$

Verify that, $w_{xx} + w_{yy} = 0$

Question no 34:-

$$f(x,y) = \tan^{-1}\left(\frac{y}{x}\right)$$

let;

$$w = \tan^{-1} \frac{y}{x}$$

$$w_x = \frac{x^2}{(x^2+y^2)} \cdot \frac{-y}{x^2} = \frac{-y}{(x^2+y^2)}$$

$$w_{xx} = \frac{-y(-2)(x)}{(x^2+y^2)^2}$$

$$w_{xx} = \frac{2yx}{(x^2+y^2)^2}$$

$$w_y = \frac{x^2}{(x^2+y^2)} \left(\frac{1}{x} \right)$$

$$w_{yy} = \frac{-2xy}{(x^2+y^2)^2}$$

$$w_{xx} + w_{yy} = \frac{2xy}{(x^2+y^2)^2} - \frac{2xy}{(x^2+y^2)^2} = 0$$

Question no 35:-

$$f(x,y) = \cos x \sinh y + \sin x \cosh y$$

$$W = \cos x \sinh y + \sin x \cosh y$$

$$W_x = -\sin x \sinh y + \cos x \cosh y$$

$$W_{xx} = -\cos x \sinh y - \sin x \cosh y$$

$$W_y = \cos x \cosh y + \sin x \sinh y$$

$$W_{yy} = \cos x \sinh y + \sin x \cosh y$$

$$W_{xx} + W_{yy} = -\cos x \sinh y - \sin x \cosh y + \cos x \sinh y + \sin x \cosh y$$

$$W_{xx} + W_{yy} = 0$$

Question no 36:-

$$f(x,y) = e^{-x} \cos y + e^{-y} \cos x$$

$$W = e^{-x} \cos y + e^{-y} \cos x$$

$$W_x = -e^{-x} \cos y + e^{-y} (-\sin x)$$

$$W_{xx} = e^{-x} \cos y - e^{-y} \cos x$$

$$W_y = -e^{-x} \sin y - e^{-y} \cos x$$

$$W_{yy} = -e^{-x} \cos y + e^{-y} \cos x$$

$$W_{xx} + W_{yy} = e^{-x} \cos y - e^{-y} \cos x - e^{-x} \cos y + e^{-y} \cos x$$

$$W_{xx} + W_{yy} = 0$$

Question no 37:-

$$W = \cos(x-y) + \ln(x+y)$$

Show that, $W_{xx} - W_{yy} = 0$

$$W_x = -\sin(x-y) + (x+y)^{-1}$$

$$W_{xx} = -\cos(x-y) - (x+y)^{-2}$$

$$W_y = +\sin(x-y) + (x+y)^{-1}$$

$$W_{yy} = -\cos(x-y) - (x+y)^{-2}$$

$$W_{xx} - W_{yy} = -\cos(x-y) - (x+y)^{-2} + \cos(x-y) + (x+y)^{-2}$$

$$W_{xx} - W_{yy} = \underline{\underline{0}}$$

Question no 38:-

$$W = (y-2x)^3 - \sqrt{y-2x}$$

Show that, $W_{xx} - 4W_{yy} = 0$

$$W = (y-2x)^3 - \sqrt{y-2x}$$

$$W_x = 3(y-2x)^2(-2) - \frac{(-2)}{2\sqrt{y-2x}}$$

$$W_{xx} = -6(y-2x)^2 + \frac{1}{\sqrt{y-2x}}$$

$$W_{xx} = 24(y-2x) + \frac{1}{(y-2x)^{\frac{3}{2}}}$$

$$W_y = 3(y-2x)^2 - \frac{1}{2\sqrt{y-2x}}$$

$$W_{yy} = 6(y-2x) + \frac{1}{4(y-2x)^{\frac{3}{2}}}$$

$$W_{xx} - 4W_{yy} = 24(y-2x) + \frac{1}{(y-2x)^{\frac{3}{2}}} - 4 \left[6(y-2x) + \frac{1}{4(y-2x)^{\frac{3}{2}}} \right]$$

$$W_{xx} - 4W_{yy} = 0 \text{ proved}$$

Question no 39:-

$$W = e^{-c^2 t} \sinh cx, \quad W_{xx} = W_t$$

$$W = e^{-c^2 t} \sinh cx$$

$$W_x = ce^{-c^2 t} \cosh cx$$

$$W_{xx} = c^2 e^{-c^2 t} \sinh cx = \frac{d}{dt} (e^{-c^2 t} \sinh cx)$$

$$W_t = -c^2 e^{-c^2 t} \sinh cx$$

$$\text{Thus, } W_{xx} = W_t$$

Question no 40:-

$$PV = knT$$

$$\frac{\partial V}{\partial T} = \frac{kn}{P}$$

$$\frac{\partial T}{\partial P} = \frac{V}{kn}$$

$$\frac{\partial P}{\partial V} = -\frac{knT}{V^2}$$

$$\frac{\partial P}{\partial V} = -\frac{knT}{V^2}$$

$$\frac{\partial P}{\partial V} = -\frac{knT}{V^2}$$

$$\frac{\partial P}{\partial V} = -\frac{knT}{V^2}$$

$$\text{Thus, } \frac{\partial V}{\partial T} \frac{\partial T}{\partial P} \frac{\partial P}{\partial V} = \frac{kn}{P} \frac{V}{kn} = \frac{V}{P} = \frac{knT}{P^2}$$

$$\frac{\partial v}{\partial T} \frac{\partial T}{\partial P} \frac{\partial P}{\partial v} = \frac{-kT}{PV} = -1, \text{ as } PV = kT$$

Show that:-

$$\frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2}$$

Question no 41:-

$$v = (\sin akt)(\sin kn)$$

$$\frac{\partial v}{\partial t} = (ak \cos(akt))(\sin kn) + 0$$

$$\frac{\partial^2 v}{\partial t^2} = -a^2 k^2 \sin(akt) \sin kn$$

$$\frac{\partial v}{\partial x} = k \sin(akt) \cos kn$$

$$\frac{\partial^2 v}{\partial x^2} = -k^2 \sin(akt) \sin(kn)$$

$$a^2 \frac{\partial^2 v}{\partial x^2} = -a^2 k^2 \sin(akt) \sin(kn)$$

∴ PROVED

Question no 42:-

$$v = (x - at)^4 + \cos(x + at)$$

$$\frac{\partial v}{\partial t} = 4(x - at)^3(-a) - a \sin(x + at)$$

$$\frac{\partial^2 v}{\partial t^2} = 12a^2(x-at)^2 - a^2 \cos(x+at)$$

$$\frac{\partial v}{\partial x}$$

$$= 4(x-at)^3 - a \sin(x+at)$$

$$\frac{\partial^2 v}{\partial x^2}$$

$$= 12(x-at)^2 - \cos(x+at)$$

$$\frac{\partial^2 v}{\partial x^2}$$

$$= 12a^2(x-at)^2 - a^2 \cos(x+at)$$

$$\frac{\partial^2 v}{\partial x^2}$$

Hence proved,

$$\frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2}$$

Show Cauchy-Riemann equations,

$$u_x = v_y \quad \text{and}$$

$$u_y = -v_x$$

Question no 43:-

$$u(x,y) = x^2 - y^2, \quad v(x,y) = 2xy$$

$$u_x = 2x$$

$$u_y = -2y$$

$$v_x = 2y$$

$$v_y = 2x$$

Satisfy the Cauchy-Riemann equation

Question no 44:-

$$u(x,y) = \frac{y}{(x^2+y^2)}, \quad v = \frac{x}{x^2+y^2}$$

$$u_x = \frac{-2xy}{(x^2+y^2)^2}$$

$$u_y = \frac{(x^2+y^2) - (2y^2)}{(x^2+y^2)^2}$$

$$u_y = \frac{x^2 - y^2}{(x^2+y^2)^2}$$

$$v_x = \frac{(x^2+y^2) - 2x^2}{(x^2+y^2)^2}$$

$$v_x = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

$$v_y = \frac{-2xy}{(x^2+y^2)^2}, \quad \text{Satisfy Cauchy-Riemann}$$

Question no 45:-

$$u(x,y) = e^x \cos y, \quad v(x,y) = e^x \sin y$$

$$u_x = e^x \cos y$$

$$u_y = -e^x \sin y$$

$$v_x = e^x \cos y \sin y$$

$$v_y = e^x \cos y$$

PROVE,

$$u_x = v_y$$

$$u_y = -v_x$$

Question no 46:-

$$u(x,y) = \cos x \cosh y + \sin x \sinh y$$

$$v(x,y) = \cos x \cosh y - \sin x \sinh y$$

$$u_x = -\sin x \cosh y + \cos x \sinh y$$

$$v_y = \cos x \sinh y - \sin x \cosh y$$

$$u_y = \cos x \sinh y + \sin x \cosh y$$

$$v_x = -\sin x \cosh y - \cos x \sinh y$$

Proved, $-v_x = u_y$

$u_x = v_y$

Question no 47:-

There are 4 second partial derivative:

$$W_{xx}, W_{xy}, W_{xz}, W_{yx}, W_{yy}, W_{yz}, W_{zx}, W_{zy}, W_{zz}$$

Question no 48:-

$w = f(x, y, z, t, v)$, define w_t as a limit.

$$w_t = \lim_{h \rightarrow 0} \frac{f(x, y, z, t+h, v) - f(x, y, z, t, v)}{h}$$

Question no 49:-

If f has continuous third partial derivatives, ten of them would be different.

Exercise 16.5

Use chain rule in ques 1 to 14:-

Question no 1:-

$$W = u \sin v, \quad u = x^2 + y^2, \quad v = xy$$

$$\frac{\partial W}{\partial x} = \frac{\partial W}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial W}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= \sin v (2x) + u \cos v (y)$$

$$= 2x \sin(xy) + y(x^2 + y^2) \cos(xy)$$

$$\frac{\partial W}{\partial y} = \frac{\partial W}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial W}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= \sin v (2y) + u \cos v (x)$$

$$= 2y \sin xy + (x^2 + y^2)x \cos xy$$

Question no 2:-

$$W = uv + v^2, \quad u = x \sin y, \quad v = y \sin x$$

$$\frac{\partial W}{\partial x} = \frac{\partial W}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial W}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= v \sin y + (u + 2v)(y \cos x)$$

$$= y \sin x \sin y + (x \sin y + 2y \sin x)(y \cos x)$$

$$\frac{\partial W}{\partial y} = \frac{\partial W}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial W}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= v \sin x \cos y + y \cos x (\sin x)(u + 2v)$$

$$= xy \sin x \cos y + y(x \sin y + 2y \sin x)(\sin x)$$